

Create a Safe Google Cloud Training Environment

75 minutes · Mapped criteria A1, A2, K1, K3

Purpose and work product

Create an account, project, billing guardrails and CLI configuration, then prove the environment is safe to use.

Submit: environment-check.txt, budget-policy.json and a project inventory

Safety, cost and privacy

Use only the assigned training project. Confirm the active account, project and region before every mutation. Never place credentials, payment details, access tokens or private keys in screenshots or submitted files. Stop billable resources and verify the final inventory.

Files in this folder

File	Purpose
check_environment.sh	Mock data, script or configuration used by this activity
budget-policy.json	Mock data, script or configuration used by this activity

Detailed procedure

Step 1

Open <https://cloud.google.com/> and select Get started for free. Sign in with a Google Account you control. Create one first if required, then enable multi-factor authentication in the Google Account security settings.

Step 2

Review the current trial eligibility, country, payment and automatic-upgrade terms. Do not use a corporate billing account unless your organisation has approved the training spend.

Step 3

Accept the Google Cloud terms. If prompted, create a Cloud Billing account and add the required payment method. Record only the billing account name; never place card details in an activity submission.

Step 4

Open <https://console.cloud.google.com/>. Use the project selector at the top and choose New Project. Name it ace-training-<your initials>. Copy the generated project ID exactly; the display name and project ID are different.

Step 5

From Billing, link only the new training project to the training billing account. Verify the Billing overview names the same project ID.

Step 6

From Billing > Budgets & alerts, create ACE training budget. Set a deliberately small amount agreed by the trainer and add thresholds at 50%, 90% and 100%. Remember that a budget alerts but does not stop spending.

Step 7

Open Cloud Shell from the console header. Wait for the terminal prompt, then run gcloud auth list. Confirm the active identity is yours.

Step 8

Set PROJECT_ID to the copied project ID, export it for this shell, and run gcloud config set project \$PROJECT_ID. Run gcloud config list and confirm both account and project.

Step 9

Enable the Compute Engine, Service Usage, Cloud Logging and Cloud Monitoring APIs. Wait for the operation to finish and list enabled services.

Step 10

Run the supplied check_environment.sh. Save its output as environment-check.txt. Capture the project ID, billing enabled state, enabled APIs and timestamp.

Step 11

Negative test: switch the script to a deliberately invalid project ID and record the error. Restore the correct project immediately. Explain how this test prevents accidental work in the wrong project.

Step 12

Open Billing reports and record the current cost baseline. Remove any test resource created accidentally. Keep the project for subsequent activities.

Verification record

Record expected and observed output. A successful command without the correct project, identity, scope or evidence is not an accepted result.

Check	Expected evidence
Identity and scope	Active account, project ID, region or zone
Configuration	Resource description or policy output
Runtime result	Test output with UTC timestamp
Failure test	One controlled negative test and diagnosis
Cleanup	Final inventory and stopped/deleted billable resources

Troubleshooting sequence

Check the active identity and project first, then API enablement and IAM, then resource state, networking and quotas. Preserve the failed output before making a correction. If the issue remains, escalate with the exact command, timestamp, expected result, observed result and evidence files.

Cleanup

Delete or stop resources as instructed. Re-open Billing and the relevant service inventory, then record the remaining resource names. Budget alerts do not stop charges.